

Kubernetes Assignment

1. Kubernetes and Its Architecture

Kubernetes (K8s) is an open-source container orchestration platform used to automate the deployment, scaling, and management of containerized applications. It helps organizations run applications reliably across multiple servers while ensuring high availability and efficient resource utilization.

Architecture:

The Kubernetes architecture consists of two major components:

Control Plane (Master Node) – Responsible for managing the cluster. It includes: **API Server:** Entry point for all Kubernetes commands. **Scheduler:** Assigns Pods to suitable worker nodes. **Controller Manager:** Maintains the desired state of the cluster. **etcd:** Distributed key-value store used to store cluster data. **Worker Nodes** – Execute application workloads. Each node contains: **Kubelet:** Communicates with the control plane and manages Pods. **Container Runtime:** Runs containers (e.g., containerd). **Kube Proxy:** Handles networking and service communication. Workflow: Users send requests to the API Server, the Scheduler selects a node, Kubelet creates Pods, and Services expose applications to users.

2. Nodes, Pods, Services, Deployments, and ReplicaSets

Node

A Node is a physical or virtual machine that runs application workloads. Worker nodes provide CPU, memory, storage, and networking resources for Pods.

Pod

A Pod is the smallest deployable unit in Kubernetes. It contains one or more containers that share the same network and storage resources. Pods are temporary and can be recreated if they fail.

Service

A Service provides a stable network endpoint for accessing Pods. Since Pod IP addresses can change, Services ensure consistent communication between applications and users.

Deployment

A Deployment is used to manage application updates and scaling. It defines the desired number of Pod instances and supports rolling updates, rollbacks, and automated recovery.

ReplicaSet

A ReplicaSet ensures that a specified number of identical Pods are always running. If a Pod fails, the ReplicaSet automatically creates a replacement. Deployments use ReplicaSets internally for Pod management.

Conclusion

Kubernetes simplifies container management through automation, scalability, and self-healing features. Nodes provide infrastructure, Pods run applications, Services enable communication, Deployments manage updates, and ReplicaSets maintain application availability.